A Transport–Spectral Contradiction Program for the Riemann Hypothesis

Closure, Explosive Obstruction, Common Readout Channels, the Parity Bridge Program, and Addenda A/B/C/D

Abstract

We assemble the current transport–spectral contradiction framework for the Riemann Hypothesis in a single coherent document. The closure side is formulated in terms of an exact window law for a dipole-regularized explicit residue channel, implying vanishing excess. The obstruction side is formulated in terms of an off-critical lifted drift variable whose quartic penalty is coercively unbounded. Under a uniform bridge inequality from the off-critical mode to the witness bank, the normalized excess diverges to $+\infty$, yielding contradiction with closure. We also incorporate an exact common-readout channel theorem, showing that distinct channels may be readable through a shared terminal sector without collapse of distinction. Finally, we record the parity oscillator, odd-parity projection, Catalan block, and parity-weighted mini-zeta channel as the concrete bridge program. The resulting manuscript is mathematically honest: it gives a sharp conditional contradiction theorem and a structured channel architecture for the missing bridge, but does not yet claim the unconditional proof of the Riemann Hypothesis.

1 Introduction

The intended contradiction has the form

$$E_{\mathrm{ex}}(f) = 0 \qquad \text{and} \qquad \exists \rho \text{ off-critical} \Longrightarrow E_{\mathrm{ex}}(f) = +\infty.$$

If both statements are proved for the same explicit residue channel f, then the Riemann Hypothesis follows.

The present document has four goals:

(i) isolate the already clean closure theorem;

(ii) isolate the already clean explosive-obstruction theorem under a bridge input;

(iii) record the exact common-readout channel theorem, showing how distinct channels may share a terminal readout sector without becoming identical;

(iv) formulate the parity/Catalan/mini-zeta mechanism as the concrete transport-to-bank bridge program.

The guiding principle is honesty: no step is promoted to an unconditional theorem unless it is explicitly proved in the text.

2 Residue channel, witness bank, and excess

Definition 2.1 (Explicit residue distribution)**.** Let μ denote an explicit-formula residue distribution on $\mathbb{R}$ in a Guinand–Weil normalization. On the Fourier side, we write schematically

$$\widehat{\mu}(\omega) = \sum_{n\geq 1} \frac{\Lambda(n)}{\sqrt{n}}\, \delta(\omega - \log n) + G(\omega),$$

where G is the archimedean contribution.

Definition 2.2 (Dipole-regularized residue channel)**.** Fix a Schwartz atom $\psi \in \mathscr{S}(\mathbb{R})$ satisfying the dipole condition

$$\widehat{\psi}(0) = 0,$$

and define

$$f := \psi * \mu.$$

Definition 2.3 (Witness bank)**.** Fix dyadic scales

$$a_j = 2^j a_0, \qquad j = 0, \dots, J,$$

and positive weights $w_j > 0$. For any channel h, define the second-difference witness

$$(W_a h)(\tau) := h(\tau + a) - 2h(\tau) + h(\tau - a).$$

Let $\varphi_{\tau,L}$ be a nonnegative Schwartz window localized on $[\tau, \tau + L]$. Define the bank energy

$$Q_{\text{bank}}(h; \tau, L) := \sum_{j=0}^{J} w_j^2 \, \langle W_{a_j} h, \varphi_{\tau,L} W_{a_j} h \rangle.$$

Definition 2.4 (Excess functional)**.** Let $E_0 \geq 0$ be the baseline density. Define

$$E_{\text{ex}}(f) := \limsup_{L \to \infty} \sup_{\tau \in \mathbb{R}} \frac{Q_{\text{bank}}(f; \tau, L) - E_0 L}{L}.$$

3 Closure: exact window law and vanishing excess

Definition 3.1 (Exact window law)**.** We say that the explicit residue channel f satisfies the exact window law if there exist a bounded function

$$J_f : \mathbb{R} \to \mathbb{R}$$

and a remainder $R(\tau, L)$ such that

$$Q_{\text{bank}}(f; \tau, L) = E_0 L + J_f(\tau + L) - J_f(\tau) + R(\tau, L),$$

with

$$\sup_{\tau \in \mathbb{R}} |J_f(\tau)| < \infty, \qquad \lim_{L \to \infty} \sup_{\tau \in \mathbb{R}} \frac{|R(\tau, L)|}{L} = 0.$$

Theorem 3.2 (Vanishing excess under the exact window law)**.** *If the exact window law holds, then*

$$E_{\text{ex}}(f) = 0.$$

Proof. Divide the exact window law by L:

$$\frac{Q_{\text{bank}}(f; \tau, L) - E_0 L}{L} = \frac{J_f(\tau + L) - J_f(\tau)}{L} + \frac{R(\tau, L)}{L}.$$

Taking absolute values and the supremum over τ gives

$$\sup_{\tau \in \mathbb{R}} \left| \frac{Q_{\text{bank}}(f; \tau, L) - E_0 L}{L} \right| \leq \sup_{\tau \in \mathbb{R}} \left| \frac{J_f(\tau + L) - J_f(\tau)}{L} \right| + \sup_{\tau \in \mathbb{R}} \frac{|R(\tau, L)|}{L}.$$

Since J_f is bounded,

$$\sup_{\tau\in\mathbb{R}}\left|\frac{J_f(\tau+L)-J_f(\tau)}{L}\right| \le \frac{2\sup_\tau |J_f(\tau)|}{L} \longrightarrow 0.$$

By assumption,

$$\sup_{\tau\in\mathbb{R}}\frac{|R(\tau,L)|}{L} \longrightarrow 0.$$

Therefore

$$\limsup_{L\to\infty}\sup_{\tau\in\mathbb{R}}\frac{Q_{\mathrm{bank}}(f;\tau,L)-E_0L}{L}=0,$$

that is,

$$E_{\mathrm{ex}}(f)=0.$$

□

4 Off-critical lifted drift and quartic obstruction

Definition 4.1 (Lifted drift)**.** Let

$$\rho=\beta+i\gamma$$

be a nontrivial zero of $\zeta(s)$. In log-time $\tau=\log t$, define the lifted drift

$$\delta_\rho(\tau):=\frac{\beta-\frac{1}{2}}{2\pi}\tau+r_\rho(\tau),$$

where r_ρ is a bounded remainder term.

Remark 4.2. The quantity δ_ρ must be treated as a lifted *unwrapped* variable. A bounded periodic surrogate such as $\sin^4(\delta_\rho(\tau))$ is not coercive enough for the contradiction mechanism.

Definition 4.3 (Quartic drift penalty)**.** For an off-critical mode ρ, define

$$P_4(\rho;\tau,L):=\int_\tau^{\tau+L}|\delta_\rho(u)|^4\,du.$$

Lemma 4.4 (Uniform quartic window lower bound)**.** *Assume*

$$\delta_\rho(u)=au+b(u),\qquad a\neq 0,\qquad |b(u)|\le M\quad\forall u\in\mathbb{R}.$$

Then for every

$$\tau\ge\tau_0:=\frac{2M}{|a|}$$

and every $L>0$, one has

$$P_4(\rho;\tau,L)\ge\frac{|a|^4}{16}\int_\tau^{\tau+L}u^4\,du.$$

Hence

$$P_4(\rho;\tau,L)\ge\frac{|a|^4}{16}\left(\tau^4L+2\tau^3L^2+2\tau^2L^3+\tau L^4+\frac{1}{5}L^5\right).$$

Proof. Since $|b(u)|\le M$,

$$|\delta_\rho(u)|=|au+b(u)|\ge|a|u-M.$$

If $u\ge\tau\ge 2M/|a|$, then

$$M\le\frac{|a|u}{2},$$

hence

$$|a|u - M \geq \frac{|a|u}{2}.$$

Therefore

$$|\delta_\rho(u)|^4 \geq \frac{|a|^4}{16} u^4 \qquad (u \in [\tau, \tau + L]).$$

Integrating gives

$$P_4(\rho; \tau, L) \geq \frac{|a|^4}{16} \int_\tau^{\tau+L} u^4 \, du.$$

Finally,

$$\int_\tau^{\tau+L} u^4 \, du = \frac{(\tau + L)^5 - \tau^5}{5} = \tau^4 L + 2\tau^3 L^2 + 2\tau^2 L^3 + \tau L^4 + \frac{1}{5} L^5.$$

□

5 Uniform bridge implies explosive excess

Theorem 5.1 (Explosive excess from a surviving linear drift)**.** *Assume that for some off-critical component ρ,*

$$\delta_\rho(u) = au + b(u), \qquad a \neq 0, \qquad |b(u)| \leq M \quad \forall u,$$

and let

$$P_4(\rho; \tau, L) := \int_\tau^{\tau+L} |\delta_\rho(u)|^4 \, du.$$

Assume moreover that there exist constants

$$c_1 > 0, \qquad c_2 \geq 0, \qquad \tau_1 \in \mathbb{R}, \qquad L_1 \geq 1$$

such that for all

$$\tau \geq \tau_1, \qquad L \geq L_1,$$

the uniform bridge inequality holds:

$$Q_{\text{bank}}(f; \tau, L) - E_0 L \geq c_1 P_4(\rho; \tau, L) - c_2 L.$$

Then

$$E_{\text{ex}}(f) = +\infty.$$

Proof. Choose

$$\tau \geq \tau_* := \max\left\{\tau_1, \frac{2M}{|a|}\right\}.$$

By the previous lemma,

$$P_4(\rho; \tau, L) \geq \frac{|a|^4}{16} \left(\tau^4 L + 2\tau^3 L^2 + 2\tau^2 L^3 + \tau L^4 + \frac{1}{5} L^5\right)$$

for every $L > 0$. Substituting into the bridge inequality yields

$$Q_{\text{bank}}(f; \tau, L) - E_0 L \geq \frac{c_1 |a|^4}{16} \left(\tau^4 L + 2\tau^3 L^2 + 2\tau^2 L^3 + \tau L^4 + \frac{1}{5} L^5\right) - c_2 L.$$

Dividing by L gives

$$\frac{Q_{\text{bank}}(f; \tau, L) - E_0 L}{L} \geq \frac{c_1 |a|^4}{16} \left(\tau^4 + 2\tau^3 L + 2\tau^2 L^2 + \tau L^3 + \frac{1}{5} L^4\right) - c_2.$$

For every fixed $\tau \geq \tau_*$, the right-hand side tends to $+\infty$ as $L \to \infty$ because of the positive quartic term

$$\frac{c_1|a|^4}{80}L^4.$$

Therefore

$$\sup_{\tau\in\mathbb{R}} \frac{Q_{\text{bank}}(f;\tau,L) - E_0 L}{L} \to +\infty,$$

and hence

$$E_{\text{ex}}(f) = +\infty.$$

□

Corollary 5.2 (One-line form)**.** *If a genuine linear drift survives into a uniform bridge inequality, then the excess does not merely become positive: it diverges,*

$$E_{\text{ex}}(f) = +\infty.$$

6 Conditional contradiction theorem

Theorem 6.1 (Closure plus bridge implies RH)**.** *Assume:*

(1) *the exact window law holds for the explicit residue channel $f = \psi * \mu$;*

(2) *for every off-critical zero, the lifted drift law and the uniform bridge inequality hold.*

Then the Riemann Hypothesis holds.

Proof. By the closure theorem,

$$E_{\text{ex}}(f) = 0.$$

Assume for contradiction that an off-critical zero exists. Then the previous theorem gives

$$E_{\text{ex}}(f) = +\infty,$$

which contradicts $E_{\text{ex}}(f) = 0$. Hence no off-critical zero exists, so every nontrivial zero satisfies

$$\Re(s) = \frac{1}{2}.$$

□

7 Common readout sector without collapse of distinction

Lemma 7.1 (Complex substitution over derivation)**.** *Let $\Omega, \Omega' \subset \mathbb{C}$ be open sets, let*

$$u : \Omega \to \Omega', \qquad u = u(z,\bar{z}) \in C^1(\Omega),$$

and let

$$F = F(w,\bar{w}) \in C^1(\Omega').$$

Define the substitution operator

$$(U_u F)(z,\bar{z}) := F\big(u(z,\bar{z}), \overline{u(z,\bar{z})}\big).$$

Then one has the exact Wirtinger chain-rule identity

$$\begin{pmatrix} \partial_z \\ \partial_{\bar{z}} \end{pmatrix} \circ U_u = \begin{pmatrix} M_{u_z} & M_{(\bar{u})_z} \\ M_{u_{\bar{z}}} & M_{(\bar{u})_{\bar{z}}} \end{pmatrix} \circ \begin{pmatrix} U_u & 0 \\ 0 & U_u \end{pmatrix} \circ \begin{pmatrix} \partial_w \\ \partial_{\bar{w}} \end{pmatrix},$$

where M_g denotes multiplication by the function g.

Proof. By definition,

$$(U_u F)(z,\bar{z}) = F\big(u(z,\bar{z}), \overline{u(z,\bar{z})}\big).$$

Applying ∂_z and $\partial_{\bar{z}}$ gives

$$\partial_z(U_u F) = U_u(F_w)\, u_z + U_u(F_{\bar{w}})\, (\bar{u})_z,$$

and

$$\partial_{\bar{z}}(U_u F) = U_u(F_w)\, u_{\bar{z}} + U_u(F_{\bar{w}})\, (\bar{u})_{\bar{z}}.$$

Collecting these identities in vector form yields the stated formula. □

Theorem 7.2 (Common readout sector without collapse of distinction)**.** *Let $\Omega, \Omega' \subset \mathbb{C}$ be open, let A be a nonempty index set, and let*

$$u_\alpha : \Omega \to \Omega', \qquad u_\alpha \in C^1(\Omega), \qquad \alpha \in A,$$

be a family of channels. For each $\alpha \in A$, define

$$(U_\alpha F)(z,\bar{z}) := F\big(u_\alpha(z,\bar{z}), \overline{u_\alpha(z,\bar{z})}\big).$$

Let Y be a realization space and let

$$Q : \Omega' \to \mathcal{P}(Y)$$

be a set-valued terminal readout map. Assume:

(1) *the family $\{u_\alpha\}_{\alpha\in A}$ is pairwise distinct as maps;*

(2) *for each $\alpha \in A$ there exists a point $z_\alpha \in \Omega$;*

(3) *there exists a nonempty set $B \subseteq Y$ such that*

$$B \subseteq Q\big(u_\alpha(z_\alpha)\big) \qquad \text{for all } \alpha \in A.$$

Then:

(1) *each channel satisfies the exact differential transport law*

$$\begin{pmatrix} \partial_z \\ \partial_{\bar{z}} \end{pmatrix} \circ U_\alpha = \begin{pmatrix} M_{(u_\alpha)_z} & M_{(\overline{u_\alpha})_z} \\ M_{(u_\alpha)_{\bar{z}}} & M_{(\overline{u_\alpha})_{\bar{z}}} \end{pmatrix} \circ \begin{pmatrix} U_\alpha & 0 \\ 0 & U_\alpha \end{pmatrix} \circ \begin{pmatrix} \partial_w \\ \partial_{\bar{w}} \end{pmatrix};$$

(2) *the terminal readouts have a common nontrivial sector:*

$$\bigcap_{\alpha\in A} Q\big(u_\alpha(z_\alpha)\big) \neq \varnothing;$$

(3) *the existence of this common sector is compatible with pairwise distinction of the channels;*

(4) *consequently, distinct channels may be readable through a shared terminal sector without becoming identical.*

Proof. Part **(1)** follows by applying the previous lemma to $u = u_\alpha$.

For part **(2)**, assumption **(3)** gives

$$B \subseteq Q\big(u_\alpha(z_\alpha)\big) \qquad \forall \alpha \in A.$$

Therefore

$$B \subseteq \bigcap_{\alpha \in A} Q\big(u_\alpha(z_\alpha)\big).$$

Since $B \neq \varnothing$, it follows that

$$\bigcap_{\alpha \in A} Q\big(u_\alpha(z_\alpha)\big) \neq \varnothing.$$

Part **(3)** is immediate from combining assumption **(1)** with the conclusion of **(2)**.

Part **(4)** is exactly the interpretation of **(1)**–**(3)**. □

Corollary 7.3 (Holomorphic reduction)**.** *Under the assumptions of the theorem, fix $\alpha \in A$. If*

$$(u_\alpha)_{\bar{z}} = 0 \qquad \textit{and} \qquad F_{\bar{w}} = 0,$$

then the differential transport law reduces to the holomorphic channel

$$\partial_z \circ U_\alpha = M_{(u_\alpha)_z} \circ U_\alpha \circ \partial_w.$$

Proof. All antiholomorphic and mixed terms vanish in the matrix identity. □

Remark 7.4 (Generator–observer specialization)**.** The generator–observer picture is recovered by choosing index sets I and J, families

$$G_i : \Omega \to \Omega, \qquad O_j : \Omega \to \Omega',$$

and defining

$$u_{ij} := O_j \circ G_i, \qquad (i,j) \in I \times J.$$

Then the channel index set is $A = I \times J$. The two-channel, four-channel, and n-channel cases are all instances of the same formal machine. What changes is not the transport law itself, but only the size and combinatorics of the index set.

Remark 7.5 (Principle)**.** Unity of readout does not force identity of source. A common terminal sector means only that several distinct channels are readable in a shared readout regime. It does not mean that the underlying channels have become equal.

8 Readout-frame transport, iterated pullback, and terminal overlap

We now add a second channel-theoretic layer clarifying the relation between source distinction, frame transport, and terminal overlap. Its role in the bridge program is to separate three phenomena that should never be conflated:

(1) distinction of source generators,

(2) transport of the effective law across representation frames,

(3) overlap at the terminal set-valued readout layer.

Lemma 8.1 (Readout-frame minilemma)**.** *Let X be a set of generators, Y a realization space, and*

$$U : X \to \mathcal{P}(Y)$$

a set-valued readout/realization map. Let

$$E_-, E_+ \in X, \qquad E_- \neq E_+.$$

Let $u : \Omega \to \Omega'$ be a C^1-map and $F \in C^1(\Omega')$. Then:

(1) *distinction of generators is not erased by overlap of readouts: in general,*

$$U(E_-) \cap U(E_+) \neq \varnothing$$

does not imply

$$E_- = E_+;$$

(2) *the effective law on the readout layer transports by pullback:*

$$d(F \circ u) = u^*(dF);$$

(3) *in a nonholomorphic complex frame the same principle splits into two channels:*

$$\partial_z(U_u F) = U_u(F_w)u_z + U_u(F_{\bar{w}})\overline{u}_z,$$

$$\partial_{\bar{z}}(U_u F) = U_u(F_w)u_{\bar{z}} + U_u(F_{\bar{w}})\overline{u}_{\bar{z}}.$$

Thus a change of readout frame does not rewrite the source object; it only transports the law into a new representation.

Proof. For **(1)**, distinct hard generators need not have disjoint set-valued readouts. For example, let

$$X = \{-1, +1\}, \qquad Y = \{0, 1\},$$

and define

$$U(-1) = \{0\}, \qquad U(+1) = \{0, 1\}.$$

Then

$$-1 \neq +1,$$

but

$$U(-1) \cap U(+1) = \{0\} \neq \varnothing.$$

Hence overlap of realizations does not imply identity of generators.

For **(2)**, this is the standard pullback identity for differentials:

$$d(F \circ u) = u^*(dF).$$

For **(3)**, write

$$dF = F_w\, dw + F_{\bar{w}}\, d\bar{w}.$$

After composition with $u(z, \bar{z})$, one gets

$$d(F \circ u) = U_u(F_w)\, du + U_u(F_{\bar{w}})\, d\bar{u}.$$

Now decompose

$$du = u_z\, dz + u_{\bar{z}}\, d\bar{z}, \qquad d\bar{u} = \overline{u}_z\, dz + \overline{u}_{\bar{z}}\, d\bar{z},$$

and compare coefficients of dz and $d\bar{z}$. This yields exactly the two Wirtinger transport formulas above. □

Remark 8.2 (Compressed principle)**.** Generator distinction is not the same thing as readout separation, and law in a new frame is transported law:

$$d(F \circ u) = u^*(dF).$$

Thus an apparent collapse may be only a readout-level phenomenon, not a collapse of the source.

Corollary 8.3 (Spectral compression after frame transport)**.** *If the transported readout is separable and admits a convergent logarithmic expansion, then a higher-dimensional readout can reduce to a one-dimensional moment series*

$$I_n[K] = \sum_{m=1}^{\infty} \frac{(-1)^{m+1}}{m(m+1)^{n-1}} \mu_m[K], \qquad \mu_m[K] := \int_0^1 x^m K(x)\, dx.$$

Theorem 8.4 (Iterated readout unification)**.** *Let*

$$X_0 \xrightarrow{u_1} X_1 \xrightarrow{u_2} \cdots \xrightarrow{u_n} X_n$$

be a chain of C^1-representation maps, and let

$$F \in C^1(X_n)$$

be an observable law on the terminal frame. Assume also that the terminal readout is given by a set-valued map

$$U : X_n \to \mathcal{P}(Y),$$

where Y is the realization/output space. Let

$$E_-, E_+ \in X_0, \qquad E_- \neq E_+.$$

Then:

(1) *the effective law on the source frame is the iterated pullback*

$$d(F \circ u_n \circ \cdots \circ u_1) = u_1^* u_2^* \cdots u_n^*(dF);$$

(2) *if each u_k is injective on the relevant source sector, then source distinction is preserved through the whole hard transport chain:*

$$u_n \circ \cdots \circ u_1(E_-) \neq u_n \circ \cdots \circ u_1(E_+);$$

(3) *nevertheless, at the terminal set-valued readout layer one may still have overlap:*

$$U(u_n \circ \cdots \circ u_1(E_-)) \cap U(u_n \circ \cdots \circ u_1(E_+)) \neq \varnothing,$$

without implying

$$E_- = E_+.$$

Thus iterated frame transport preserves the source law and the source distinction, while overlap may arise only at the terminal readout layer.

Proof. For **(1)**, apply the pullback identity repeatedly:

$$d(F \circ u_n) = u_n^*(dF),$$

hence

$$d(F \circ u_n \circ u_{n-1}) = u_{n-1}^*\, d(F \circ u_n) = u_{n-1}^* u_n^*(dF).$$

Continuing inductively gives

$$d(F \circ u_n \circ \cdots \circ u_1) = u_1^* u_2^* \cdots u_n^*(dF).$$

For **(2)**, injectivity of each u_k implies that distinct points remain distinct after each step. Induction on the chain length gives

$$u_n \circ \cdots \circ u_1(E_-) \neq u_n \circ \cdots \circ u_1(E_+).$$

For **(3)**, the final statement concerns only the set-valued readout map U. Distinct points in X_n may have overlapping images in $\mathcal{P}(Y)$, since set-valued maps are not injective in general. Therefore terminal overlap does not imply identity at X_n, and hence does not imply identity back at the source. □

Corollary 8.5 (Overlap is terminal, not generative)**.** *In the setting above, any nontrivial overlap*

$$U(u_n \circ \cdots \circ u_1(E_-)) \cap U(u_n \circ \cdots \circ u_1(E_+)) \neq \varnothing$$

is a terminal readout phenomenon. It cannot be used to identify the source generators as long as the hard transport chain remains injective.

Remark 8.6 (Iterated nonholomorphic transport)**.** If some of the transport maps u_k are nonholomorphic, then at each stage the pulled-back law splits into w- and $\bar{w}$-channels. The iterated transport remains valid, but the effective law accumulates channel-mixing terms according to the Wirtinger decomposition.

9 The parity oscillator and the Catalan–mini-zeta channel

Definition 9.1 (Parity oscillator)**.** Define the discrete dynamical system

$$x_{n+1} = -x_n, \qquad x_0 = 1.$$

Lemma 9.2. *The unique solution is*

$$x_n = (-1)^n.$$

Proof. Immediate by induction. □

Definition 9.3 (Absolute energy)**.** Define

$$E_n := |x_n|.$$

Theorem 9.4 (Energy invariance)**.** *For all $n \in \mathbb{N}$,*

$$E_n = 1.$$

Proof. Since $x_n = (-1)^n$, we have $|x_n| = 1$ for all n. □

Definition 9.5 (Odd-parity projector)**.** For any sequence (a_n) define

$$(P_{\text{odd}} a)_n := \frac{1}{2}\big(1 - (-1)^n\big) a_n.$$

Lemma 9.6. *The operator P_{odd} is an idempotent projection onto the odd-index sector:*

(1) $P_{\text{odd}}^2 = P_{\text{odd}}$*;*

(2) $(P_{\text{odd}} a)_n = 0$ *for even n;*

(3) $(P_{\mathrm{odd}}a)_n = a_n$ *for odd* n.

Proof. This follows from

$$\big(1-(-1)^n\big)^2 = 2\big(1-(-1)^n\big).$$

□

Recall the classical Fourier expansions

$$\log(\sin x) = -\log 2 - \sum_{n=1}^{\infty} \frac{\cos(2nx)}{n},$$

$$\log(\cos x) = -\log 2 - \sum_{n=1}^{\infty} \frac{(-1)^n \cos(2nx)}{n}.$$

Subtracting gives

$$\log(\sin x) - \log(\cos x) = -\sum_{n=1}^{\infty} \frac{1-(-1)^n}{n} \cos(2nx).$$

Applying P_{odd} yields

$$\log(\sin x) - \log(\cos x) = -2\sum_{k=0}^{\infty} \frac{\cos(2(2k+1)x)}{2k+1}.$$

Integrating over $[0, \pi/4]$,

$$\int_0^{\pi/4} \big(\log(\sin x) - \log(\cos x)\big)\, dx = -\sum_{k=0}^{\infty} \frac{(-1)^k}{(2k+1)^2} = -G,$$

where G is the Catalan constant.

Using

$$\log\left(1+\frac{1}{n}\right) = \sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{k\, n^k},$$

define

$$S_{\mathrm{raw}} := \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n^2} \log\left(1+\frac{1}{n}\right).$$

Then

$$S_{\mathrm{raw}} = \sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{k} \big(1-2^{-(k+1)}\big)\zeta(k+2).$$

This is a parity-weighted mini-zeta Dirichlet channel.

Theorem 9.7 (Catalan–mini-zeta structural identity)**.** *Let*

$$I = \int_0^1 \int_0^1 \frac{\log(1+xy)}{1+x^2}\, dx\, dy, \qquad J = \int_0^{\pi/4} \log(1+\tan\theta)\, d\theta.$$

Then the exact identity holds:

$$I = 2J - \frac{1}{4}G - \frac{1}{4}S_{\mathrm{raw}}.$$

10 Channels integrated into the bridge program

Definition 10.1 (Single-zero parity channel)**.** Let μ_ρ denote the single-zero contribution associated with an off-critical zero

$$\rho = \beta + i\gamma, \qquad \beta \neq \frac{1}{2}.$$

Define the parity-projected channel

$$f_\rho^{\mathrm{odd}} := \psi * P_{\mathrm{odd}}(\mu_\rho).$$

Definition 10.2 (Generator–observer transport family)**.** Choose generators G_i and observers O_j, and define channels

$$u_{ij} := O_j \circ G_i, \qquad (i,j) \in I \times J.$$

Let the distinguished off-critical parity channel be one member of this family, denoted u_ρ.

Remark 10.3. The common-readout theorem together with the readout-frame/iterated-pullback results show that several distinct channels may share a terminal readout sector without collapse of identity, and that terminal overlap need not propagate backward to the source. This is exactly the formal situation one wants for the RH program: the off-critical parity channel may remain distinct while still being readable in the same terminal sector as other channels. Thus a shared readout regime does not destroy the possibility of isolating the off-critical contribution.

Lemma 10.4 (Parity survival)**.** *The projected channel f_ρ^{odd} is nontrivial and retains the off-critical coefficient carrying the factor $\beta - \frac{1}{2}$.*

Status. This is the first bridge lemma to be proved from the explicit residue-side realization of μ_ρ. The common-readout theorem supplies the channel architecture in which such survival can be stated without forcing collapse into the other channels. □

Lemma 10.5 (Channel transport to lifted drift)**.** *There exists a transported readout $T_\rho(\tau)$ associated with f_ρ^{odd} such that its unwrapped phase satisfies*

$$\Phi_\rho(\tau) - \Phi_{\mathrm{crit}}(\tau) = \frac{\beta - \frac{1}{2}}{2\pi}\tau + r_\rho(\tau), \qquad r_\rho \text{ bounded.}$$

Status. This is the second bridge lemma. The exact Wirtinger transport law from the common-readout theorem shows how a channel can be propagated through substitution without losing its differential identity; what remains is the specific large-τ asymptotic extraction that turns the off-critical parity channel into a linear lifted drift with bounded remainder. □

Lemma 10.6 (No-hiding bank lower bound)**.** *There exist constants $c_1 > 0$, $c_2 \geq 0$ such that for all sufficiently large τ and all sufficiently large L,*

$$Q_{\mathrm{bank}}(f;\tau,L) - E_0 L \geq c_1 P_4(\rho;\tau,L) - c_2 L.$$

Status. This is the decisive transport-to-bank lemma. It must show that the parity-projected off-critical channel injects a coercive positive contribution into the witness bank and that the remainder channel cannot hide it. The common-readout theorem justifies that shared readout does not collapse distinct channels; the missing point is the explicit quantitative lower bound with positive coefficient. □

Theorem 10.7 (Integrated bridge theorem)**.** *If the three bridge lemmas above hold, then every off-critical zero forces*

$$E_{\mathrm{ex}}(f) = +\infty.$$

Combined with closure, this yields the Riemann Hypothesis.

Proof. The second and third bridge lemmas supply the hypotheses of the explosive-obstruction theorem. Hence any off-critical zero implies

$$E_{\mathrm{ex}}(f) = +\infty.$$

The closure theorem gives

$$E_{\mathrm{ex}}(f) = 0.$$

Contradiction. □

11 Witness invariance and representative freedom

We now record a second model-theoretic principle: in some formulations the primary object is not the geometric representative itself, but an invariant witness attached to it. In that setting, threshold statements are witness-locked rather than line-locked.

Definition 11.1 (Witness and admissible representative shift)**.** Let $\Omega \subset \mathbb{C}$ and let

$$W : \Omega \to Y$$

be a witness with values in a readout space Y. Let

$$\Pi : \Omega \to \Omega$$

be a transformation. We say that Π is W-admissible if

$$W(\Pi(s)) = W(s) \qquad \text{for all } s \in \Omega.$$

This induces the equivalence relation

$$s \sim_W t \iff W(s) = W(t),$$

and the associated representative class

$$[s]_W := \{\Pi(s) : \Pi \text{ is } W\text{-admissible}\}.$$

Proposition 11.2 (Representative freedom)**.** *Let $W : \Omega \to Y$ be a witness, and assume that Π is W-admissible. Then any statement depending only on $W(s)$ is constant on the orbit $[s]_W$. In particular, if a threshold functional*

$$\Theta : Y \to \mathbb{R}$$

is evaluated through W, then

$$\Theta(W(\Pi(s))) = \Theta(W(s))$$

for every admissible representative shift Π.

Proof. By admissibility,

$$W(\Pi(s)) = W(s).$$

Applying Θ to both sides gives

$$\Theta(W(\Pi(s))) = \Theta(W(s)).$$

Hence every statement formulated purely in terms of W or a function of W is invariant under admissible representative changes. □

Definition 11.3 (Threshold witness)**.** Suppose relevance is decided by the scalar witness level

$$T(s) := \Theta(W(s)).$$

Fix a threshold value $T_c \in \mathbb{R}$. The decision rule is

$$T(s) \geq T_c.$$

Proposition 11.4 (Threshold relevance is class-level)**.** *If Π is W-admissible, then*

$$T(\Pi(s)) = T(s).$$

Hence

$$T(s) \geq T_c \iff T(\Pi(s)) \geq T_c.$$

Therefore threshold relevance is a property of the equivalence class $[s]_W$, not of the particular representative s.

Proof. This is immediate from the previous proposition with $T = \Theta \circ W$. □

Remark 11.5 (Location-locked versus witness-locked)**.** In the classical formulation of RH, the geometric location itself is the statement:

$$\Re(s) = \frac{1}{2}.$$

There, location is primary. In an invariant witness setting, location need not be primary. The primary object is the witness W, and admissible changes of representative are allowed provided they preserve that witness. Then the operative logic is:

$$\text{standard setting: location is the statement,}$$

whereas

$$\text{witness setting: the witness is the statement.}$$

Remark 11.6 (Role inside the bridge program)**.** This principle does not replace the classical RH statement. Rather, it clarifies a complementary mode of reading the bridge architecture: one may regard the decisive object either as a geometric representative or as an invariant witness class, provided the model is genuinely witness-locked. This is exactly the sense in which the same structure may be read "tak i tak": geometrically on the line side, or structurally on the invariant-witness side.

12 Monotone operator chains and threshold preservation

We finally record a simple but important model principle that mirrors the fusion-side "resonance ⇒ transfer ⇒ threshold" architecture. The point is not the specific physics, but the formal fact that a monotone operator chain preserves the location of a decisive peak and the truth of threshold statements.

Definition 12.1 (Monotone operator chain)**.** Let x range over a parameter space, let $L(x) \geq 0$ be a base profile, and define a chain

$$x \longmapsto L(x) \longmapsto \alpha(x) \longmapsto \Gamma(x) \longmapsto T(x),$$

where

$$\alpha(x) = \Lambda(L(x)), \qquad \Gamma(x) = G(\alpha(x)), \qquad T(x) = H(\Gamma(x)),$$

and Λ, G, H are nondecreasing on the relevant range.

Proposition 12.2 (Peak preservation through a monotone chain)**.** *If $L(x)$ attains its maximum at $x = x_*$, then the quantities*

$$\alpha(x), \qquad \Gamma(x), \qquad T(x)$$

also attain their maxima at $x = x_*$.

Proof. Since Λ is nondecreasing, $\alpha(x) = \Lambda(L(x))$ is maximal where $L(x)$ is maximal. Applying the same argument successively to G and H yields the claim. □

Proposition 12.3 (Threshold preservation)**.** *Fix a threshold T_c. If*

$$T(x_*) \geq T_c,$$

then the chain reaches threshold at the point where the base profile is maximal. Equivalently,

$$\textit{peak of } L \Longrightarrow \textit{peak of } T \Longrightarrow \textit{best chance of threshold crossing.}$$

Proof. By the previous proposition, $T(x)$ is maximal at x_*. Hence if the threshold is crossed at x_*, it is crossed at the maximally favorable point of the chain. □

Remark 12.4 (Kill-test principle)**.** Suppose a model asserts that the observable $R(x)$ is proportional to the base profile:

$$R(x) \propto L(x).$$

Then a genuine mechanism must produce a corrected signal with the same peak as the anchor mode. Thus:

$$\text{no peak} \Longrightarrow \text{no model support.}$$

This is not a proof that the mechanism is physically realized; it is a falsifiability principle.

Remark 12.5 (Role inside the bridge program)**.** Inside the RH architecture this principle serves as a formal guide: if the decisive witness is generated through a monotone chain from an off-critical source mode, then the threshold or obstruction statement should be attached to the invariant witness at the end of the chain. In that sense the bridge may be read in two compatible ways:

- geometrically, through the source-side representative;
- operationally, through the terminal witness preserved by the chain.

13 Dipole-filtered bridge layer

We record here a deterministic bridge layer based on the mirror–odd boundary channel and dipole subtraction. Its role is to isolate a low-frequency-suppressed off-critical trace before transport into the witness bank.

Definition 13.1 (Boundary odd channel and dipole subtraction)**.** Fix $0 < \varepsilon < \frac{1}{2}$ and define the boundary flux

$$S(\sigma, t) := \partial_\sigma \log |\xi(\sigma + it)|.$$

Define the mirror–odd channel

$$S_{\text{odd}}(\sigma, t) := \frac{1}{2}\big(S(\sigma, t) - S(1 - \sigma, t)\big),$$

and set

$$f(t) := S_{\text{odd}}(\varepsilon, t).$$

Let $W \in C_c^\infty(\mathbb{R})$ be even with

$$\int_{\mathbb{R}} W(u)\,du = 1, \qquad W_T(u) := T^{-1}W(u/T).$$

Define the time-smoothed trace

$$f_T(t) := (f * W_T)(t),$$

and for a local zero window

$$\Gamma(t) := \{\gamma : |\gamma - t| \le H\}, \qquad N(t) := |\Gamma(t)|,$$

define the local mean operator

$$(M_t g) := \frac{1}{N(t)} \sum_{\gamma \in \Gamma(t)} g(\gamma),$$

together with the dipole-filtered trace

$$g_T(t) := f_T(t) - M_t[f_T].$$

Then by construction

$$M_t[g_T] = 0.$$

Proposition 13.2 (Deterministic low-frequency estimate)**.** *Assume g_T is absolutely continuous and $g_T' \in L^1(\mathbb{R})$. Let*

$$\widehat{g_T}(\omega) := \int_{\mathbb{R}} g_T(t) e^{-i\omega t}\,dt.$$

Then for every $\omega \in \mathbb{R}$,

$$|\widehat{g_T}(\omega)| \le |\omega|\,\|g_T'\|_{L^1(\mathbb{R})}.$$

Consequently, for all $T \ge 1$,

$$\int_{|\omega| \le 1/T} |\widehat{g_T}(\omega)|^2\,d\omega \ll T^{-3}\|g_T'\|^2_{L^1(\mathbb{R})}.$$

Proof. Using the vanishing local DC component, write

$$\widehat{g_T}(\omega) = \int_{\mathbb{R}} g_T(t)(e^{-i\omega t} - 1)\,dt.$$

By the elementary bound $|e^{-i\omega t} - 1| \le |\omega||t|$, one gets

$$|\widehat{g_T}(\omega)| \le |\omega| \int_{\mathbb{R}} |t|\,|g_T(t)|\,dt.$$

A standard integration-by-parts estimate yields

$$\int_{\mathbb{R}} |t|\,|g_T(t)|\,dt \le \|g_T'\|_{L^1(\mathbb{R})},$$

hence the first claim. Squaring and integrating over $|\omega| \le 1/T$ gives

$$\int_{|\omega| \le 1/T} |\widehat{g_T}(\omega)|^2\,d\omega \ll \|g_T'\|^2_{L^1(\mathbb{R})} \int_{|\omega| \le 1/T} \omega^2\,d\omega \ll T^{-3}\|g_T'\|^2_{L^1(\mathbb{R})}.$$

□

Remark 13.3 (Bridge meaning)**.** This estimate is deterministic: it uses only Taylor expansion at $\omega = 0$ and does not require spacing assumptions on zeros. It shows that once one has a uniform polylogarithmic control of $\|g_T'\|_{L^1}$, the low-frequency band is polynomially suppressed. Conceptually, this is the dipole-filtered entry point into the bridge.

14 Addendum B: transport realization and bank injection

We now formalize the B/C/D package in the clearest conditional form currently available.

Definition 14.1 (Isolated off-critical transport component)**.** Let

$$\rho = \beta + i\gamma, \qquad \beta \neq \frac{1}{2},$$

be an off-critical zero. We say that ρ admits an isolated transport component if

$$f = f_\rho + f_{\text{rest}},$$

where f_ρ is a contribution associated with ρ and f_{rest} contains all remaining spectral content.

Definition 14.2 (Transport realization of the isolated mode)**.** We say that f_ρ realizes the lifted drift if there exist functions $A_\rho, \Theta_\rho : \mathbb{R} \to \mathbb{R}$ such that

$$f_\rho(\tau) = A_\rho(\tau) e^{2\pi i \delta_\rho(\tau)},$$

with

$$\delta_\rho(\tau) = \frac{\beta - \frac{1}{2}}{2\pi}\tau + r_\rho(\tau), \qquad r_\rho \text{ bounded.}$$

In particular, if

$$2\pi\delta_\rho(\tau) = \Big(\beta - \frac{1}{2}\Big)\tau + \Theta_\rho(\tau), \qquad \Theta_\rho \text{ bounded,}$$

then

$$r_\rho(\tau) = \frac{\Theta_\rho(\tau)}{2\pi}$$

is bounded.

Definition 14.3 (Witness coercivity and bank-subcritical leakage)**.** We say that the witness bank detects the lifted drift of f_ρ if there exist constants

$$\kappa > 0, \qquad C_0 \geq 0$$

such that for all sufficiently large τ,

$$\sum_{j=0}^{J} w_j^2 |W_{a_j} f_\rho(\tau)|^2 \geq \kappa |\delta_\rho(\tau)|^4 - C_0.$$

We say that the remainder channel is bank-subcritical if there exist constants

$$\eta \in [0, \kappa), \qquad C_1 \geq 0$$

such that for all sufficiently large τ,

$$\sum_{j=0}^{J} w_j^2 \Big(2\Re\big(W_{a_j} f_\rho(\tau) \overline{W_{a_j} f_{\text{rest}}(\tau)}\big) + |W_{a_j} f_{\text{rest}}(\tau)|^2\Big) \geq -\eta |\delta_\rho(\tau)|^4 - C_1.$$

Theorem 14.4 (Transport-to-bank bridge)**.** *Assume:*

(1) $f = f_\rho + f_{\text{rest}}$*;*

(2) f_ρ *realizes the lifted drift* δ_ρ*;*

(3) *the witness coercivity bound holds with constant* $\kappa > 0$*;*

(4) the remainder channel is bank-subcritical with constant $\eta < \kappa$.

Then there exist constants $c_1 > 0$ and $c_2 \geq 0$ such that for all sufficiently large τ and all $L \geq 1$,

$$Q_{\text{bank}}(f;\tau,L) - E_0 L \geq c_1 P_4(\rho;\tau,L) - c_2 L.$$

More explicitly, one may take

$$c_1 := \kappa - \eta > 0, \qquad c_2 := C_0 + C_1.$$

Proof. Expand the bank energy using

$$f = f_\rho + f_{\text{rest}} :$$

$$|W_{a_j} f|^2 = |W_{a_j} f_\rho|^2 + 2\Re\big(W_{a_j} f_\rho \overline{W_{a_j} f_{\text{rest}}}\big) + |W_{a_j} f_{\text{rest}}|^2.$$

Multiply by $w_j^2 \varphi_{\tau,L}$, sum over j, and integrate over $[\tau, \tau + L]$. By witness coercivity and bank-subcritical leakage control,

$$Q_{\text{bank}}(f;\tau,L) - E_0 L \geq (\kappa - \eta) \int_\tau^{\tau+L} |\delta_\rho(u)|^4 \, du - (C_0 + C_1) L.$$

Since

$$P_4(\rho;\tau,L) := \int_\tau^{\tau+L} |\delta_\rho(u)|^4 \, du,$$

the claim follows with

$$c_1 := \kappa - \eta, \qquad c_2 := C_0 + C_1.$$

□

Corollary 14.5 (Positive excess from an off-critical zero)**.** *Under the assumptions of the previous theorem,*

$$E_{\text{ex}}(f) > 0.$$

Proof. Divide the bridge inequality by L:

$$\frac{Q_{\text{bank}}(f;\tau,L) - E_0 L}{L} \geq c_1 \frac{P_4(\rho;\tau,L)}{L} - c_2.$$

Using the coercive growth lemma for P_4, namely

$$P_4(\rho;\tau,L) \geq c\Big|\beta - \frac{1}{2}\Big|^4 \tau^4 L - C\tau^3 L$$

for all sufficiently large τ, we obtain

$$\frac{Q_{\text{bank}}(f;\tau,L) - E_0 L}{L} \geq c_1 \Big(c\Big|\beta - \frac{1}{2}\Big|^4 \tau^4 - C\tau^3 \Big) - c_2.$$

For sufficiently large τ, the right-hand side is strictly positive. Taking $\sup_\tau$ and then $\limsup_{L\to\infty}$ yields

$$E_{\text{ex}}(f) > 0.$$

□

Theorem 14.6 (Closure plus Addendum B)**.** *Assume the exact window law*

$$Q_{\text{bank}}(f;\tau,L) = E_0 L + J_f(\tau + L) - J_f(\tau) + R(\tau,L),$$

with

$$\sup_\tau |J_f(\tau)| < \infty, \qquad \lim_{L\to\infty} \sup_\tau \frac{|R(\tau,L)|}{L} = 0.$$

Assume also that every off-critical zero admits the Addendum B hypotheses. Then the Riemann Hypothesis holds.

Proof. The exact window law implies

$$E_{\mathrm{ex}}(f) = 0.$$

If an off-critical zero existed, the previous corollary would imply

$$E_{\mathrm{ex}}(f) > 0.$$

This is a contradiction. Hence no off-critical zero exists, so every nontrivial zero lies on the critical line. □

Theorem 14.7 (Bridge Lock Theorem)**.** *Assume the bridge inequality*

$$Q_{\mathrm{bank}}(f;\tau,L) - E_0 L \geq c_1 P_4(\rho;\tau,L) - c_2 L$$

for all sufficiently large τ and all $L \geq 1$, with $c_1 > 0$, $c_2 \geq 0$. Assume moreover the coercive quartic lower bound

$$P_4(\rho;\tau,L) \geq c\left|\beta - \frac{1}{2}\right|^4 \tau^4 L - C\tau^3 L$$

for all sufficiently large τ, where $c > 0$, $C \geq 0$, and $\beta \neq \frac{1}{2}$. Set

$$A := c_1 c\left|\beta - \frac{1}{2}\right|^4, \qquad B := c_1 C, \qquad D := c_2,$$

and define

$$\tau_* := \max\left(\frac{2B}{A}, \left(\frac{2D}{A}\right)^{1/4}\right).$$

Then for every $\tau \geq \tau_*$ and every $L \geq 1$,

$$\frac{Q_{\mathrm{bank}}(f;\tau,L) - E_0 L}{L} > 0.$$

Consequently,

$$E_{\mathrm{ex}}(f) > 0.$$

Proof. Divide the bridge inequality by L:

$$\frac{Q_{\mathrm{bank}}(f;\tau,L) - E_0 L}{L} \geq c_1 \frac{P_4(\rho;\tau,L)}{L} - c_2.$$

Insert the quartic lower bound:

$$\frac{Q_{\mathrm{bank}}(f;\tau,L) - E_0 L}{L} \geq A\tau^4 - B\tau^3 - D.$$

If $\tau \geq 2B/A$, then

$$B\tau^3 \leq \frac{A}{2}\tau^4.$$

Hence

$$A\tau^4 - B\tau^3 - D \geq \frac{A}{2}\tau^4 - D.$$

If also $\tau \geq (2D/A)^{1/4}$, then

$$\frac{A}{2}\tau^4 - D \geq 0.$$

Therefore for every $\tau \geq \tau_*$,

$$A\tau^4 - B\tau^3 - D > 0,$$

so

$$\frac{Q_{\mathrm{bank}}(f;\tau,L) - E_0 L}{L} > 0$$

for all $L \geq 1$. Taking $\sup_\tau$ and then $\limsup_{L\to\infty}$ gives

$$E_{\mathrm{ex}}(f) > 0.$$

□

Remark 14.8 (B/C/D packaging)**.** Addendum B packages the real analytic heart of the remaining bridge in a concise form:

(1) spectral isolation of a bank-detectable f_ρ;

(2) transport realization of its phase law;

(3) witness coercivity for the isolated mode;

(4) no-hiding / leakage control for the remainder with $\eta < \kappa$.

Once these are available, the bridge inequality and positive excess are automatic.

15 The exact bridge tasks M1–M5

We now isolate the exact missing bridge tasks in the most economical form. The final contradiction principle is already mathematically clean; what remains is the unconditional derivation of the bridge from the explicit residue channel to the bank-level lower bound.

Definition 15.1 (The bridge task list)**.** The missing transport-to-bank bridge is decomposed into the following five tasks:

M1. Explicit single-zero extraction in the required transport normalization. One must justify a decomposition of the explicit residue channel into a distinguished off-critical single-zero component and a remainder in the exact normalization used by the transport side:

$$\mu = \mu_{\rho_0} + \mu_{\mathrm{rest}}, \qquad f = \psi * \mu = f_{\rho_0} + f_{\mathrm{rest}}.$$

M2. Parity survival of a nontrivial off-critical component. One must prove that the parity projection preserves a nontrivial off-critical witness component, namely that

$$P_{\mathrm{odd}}(\mu_{\rho_0}) \neq 0$$

in the relevant sense, and that the dipole regularization does not annihilate this component.

M3. Transport to lifted drift with bounded remainder. One must prove that the transported parity channel yields an unwrapped phase readout satisfying

$$\Phi_\rho(\tau) - \Phi_{\mathrm{crit}}(\tau) = \frac{\beta - \frac{1}{2}}{2\pi}\tau + r_\rho(\tau), \qquad r_\rho \text{ bounded.}$$

Equivalently,

$$\delta_\rho(\tau) := \Phi_\rho(\tau) - \Phi_{\mathrm{crit}}(\tau) = \frac{\beta - \frac{1}{2}}{2\pi}\tau + r_\rho(\tau).$$

M4. Witness-bank lower bound with positive coefficient. One must prove the quantitative lower bound

$$Q_{\text{bank}}(f;\tau,L) - E_0 L \geq c_1 P_4(\rho;\tau,L) - c_2 L$$

for all sufficiently large τ and all sufficiently large L, with

$$c_1 > 0, \qquad c_2 \geq 0.$$

M5. Sub-coercive cancellation of the remainder channel. One must prove that the remainder channel cannot hide the coercive contribution. Equivalently, mixed and pure-remainder terms must be shown to be sub-coercive relative to the quartic drift penalty, so that the full bank lower bound survives after recombination.

Remark 15.2 (Logical compression of M1–M5)**.** Tasks M1–M2 isolate and preserve the off-critical mode, M3 transports it into a genuine lifted drift, M4 reads that drift into bank energy with positive coefficient, and M5 prevents the remainder from cancelling the resulting coercive excess. In compressed form,

$$\text{M1–M2} \Longrightarrow \text{M3} \Longrightarrow \text{M4} \Longrightarrow \text{M5} \Longrightarrow \text{target bridge inequality.}$$

Remark 15.3 (Where the real analytic burden sits)**.** The true analytic core now sits in M3–M5. The single-zero extraction layer is essentially formal once the explicit residue splitting is frozen, and parity survival is already isolated under a non-annihilation hypothesis; the genuinely hard part is:

(1) extracting the leading transported phase term with bounded remainder;

(2) proving a bank lower bound with a strictly positive coefficient;

(3) proving that the remainder channel is only sub-coercive in the surviving traceless sector.

By the Bressan-type trace/contact elimination theorem, the scalar isotropic logarithmic mass is not the true obstruction; the surviving analytic core is traceless.

16 Final rigorous status

Theorem 16.1 (Conditional RH from closure plus bridge)**.** *Let*

$$f = \psi * \mu$$

be the dipole-regularized explicit residue channel, with

$$\widehat{\psi}(0) = 0.$$

Assume:

(1) *the exact window law holds for f, namely*

$$Q_{\text{bank}}(f;\tau,L) = E_0 L + J_f(\tau+L) - J_f(\tau) + R(\tau,L),$$

with

$$\sup_{\tau\in\mathbb{R}} |J_f(\tau)| < \infty, \qquad \lim_{L\to\infty} \sup_{\tau\in\mathbb{R}} \frac{|R(\tau,L)|}{L} = 0;$$

(2) every off-critical zero

$$\rho = \beta + i\gamma, \qquad \beta \neq \frac{1}{2},$$

yields the target bridge inequality

$$Q_{\mathrm{bank}}(f;\tau,L) - E_0 L \geq c_1 P_4(\rho;\tau,L) - c_2 L$$

for all sufficiently large τ and all $L \geq 1$, where

$$P_4(\rho;\tau,L) := \int_{\tau}^{\tau+L} |\delta_\rho(u)|^4 \, du$$

and

$$\delta_\rho(u) = \frac{\beta - \frac{1}{2}}{2\pi} u + r_\rho(u), \qquad r_\rho \text{ bounded.}$$

Then the Riemann Hypothesis holds.

Proof. By the exact window law,

$$E_{\mathrm{ex}}(f) = \limsup_{L\to\infty} \sup_{\tau\in\mathbb{R}} \frac{Q_{\mathrm{bank}}(f;\tau,L) - E_0 L}{L} = 0.$$

Assume for contradiction that there exists an off-critical zero

$$\rho = \beta + i\gamma, \qquad \beta \neq \frac{1}{2}.$$

Since r_ρ is bounded, the quartic drift penalty is coercive for large τ, hence the target bridge inequality implies

$$E_{\mathrm{ex}}(f) > 0.$$

This contradicts $E_{\mathrm{ex}}(f) = 0$. Therefore no off-critical zero exists. Hence every nontrivial zero satisfies

$$\Re(\rho) = \frac{1}{2}.$$

□

Remark 16.2 (Exact current status)**.** The contradiction principle is mathematically clean. The unresolved point is not the final contradiction itself, but the unconditional proof of the bridge from an off-critical zero to the bank-level lower bound. Equivalently, the remaining open burden is exactly the M1–M5 package, with the sharpest unresolved pressure concentrated in the B/C/D chain: transport realization, witness coercivity, and no-hiding leakage control.